# PG No. 5 $C_{2h}$ $2/m$ (b-axis setting) [ monoclinic ]

* generator : $2_{010},\ -1$
* conjugacy class

$[1]$ : $1$
$[2_{010}]$ : $2_{010}$
$[-1]$ : $-1$
$[m_{010}]$ : $m_{010}$

* symmetry operation

Table 1: Symmetry operations for 3d polar vector.

| No. | tag | matrix (polar) | det |
|---|---|---|---|
| 1 | 1 | $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ | 1 |
| 2 | $2_{010}$ | $\begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$ | 1 |
| 3 | $-1$ | $\begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$ | $-1$ |
| 4 | $m_{010}$ | $\begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ | $-1$ |